

PUFiM: A Robust and Efficient FeFET-Based Security Solution Merging Physical Unclonable Function with Compute-in-Memory for Edge AI

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Abstract—Compute-in-memory (CiM) has become a promising candidate for edge AI by reducing data movements through *in-situ* operations. However, this emerging computational paradigm also poses the vulnerability of model leakage as the weights are stored in plaintext for computing. While prior works have explored lightweight encryption methods, CiM is usually considered a separate module instead of a system component, leaving the origin of keys unclear and unprotected. Physical unclonable functions (PUFs) offer a potential origin of keys, but a comprehensive framework for securing key generation and delivery remains lacking. Besides, the complementary ciphertext storage incurs substantial costs and degrades the performance.

This work proposes PUFiM, a robust and efficient security solution for edge computing based on ferroelectric FETs (FeFETs). For the first time, a strong PUF is synergized with CiM to enable authentication, key generation, and encrypted computations within a unified array for comprehensive protection. To achieve this synergization, a high-density hybrid storage and computation approach combining PUF and weight bits via multi-level cell (MLC) FeFETs is proposed. Besides, two PUF enhancement techniques and a novel mapping scheme are developed to improve security and efficiency further. Results show that PUFiM withstands PUF modeling attacks with up to 10M samples. Moreover, PUFiM reduces the inference accuracy by >60% under 95% key leakage and achieves >9.7× compute density and >1.2× energy efficiency improvement compared with the state-of-the-art SRAM/NVM secure CiMs.

Keywords—Hardware Security, Computing-in-Memory, Physical Unclonable Functions, Ferroelectric FET (FeFET)

I. INTRODUCTION

Deep neural networks (DNNs) have shown superior performance in diverse application domains, from visual recognition to natural language processing [1]–[3]. With the increasing demand for privacy and real-time responses, DNN models are migrating from cloud to edge by deploying the trained models on edge devices for inference. However, in conventional von Neumann architectures, efficient DNN inference faces the bottleneck of substantial data movements between processing units and memories [4]. This memory wall challenge prompts the emerging computational paradigm of compute-in-memory (CiM), which aims to reduce data movements via *in-situ* operations [5], [6]. Towards the paradigm, various impressive implementations have been demonstrated, including CMOS CiMs [7], [8] and nonvolatile memory (NVM) CiMs [9]–[12].

However, in the unsupervised environment, CiMs suffer from the vulnerability of model leakage as the weights are stored in plaintext for computing. This data theft risk is

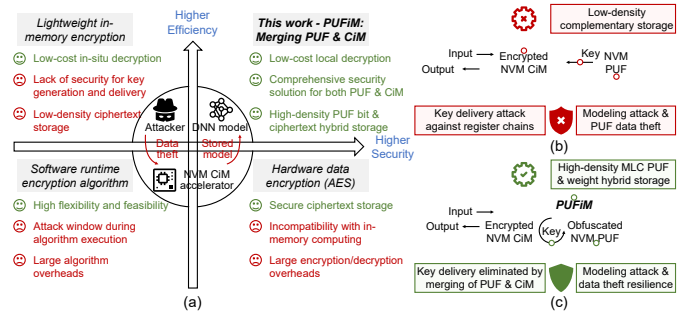


Fig. 1. Secure CiMs: (a) Comparison between encryption methods. (b) Challenges in existing designs. (c) Insights of the proposed PUFiM.

exacerbated when NVMs are employed, as shown in Fig. 1(a). Therefore, weight encryption becomes crucial to protect the DNN models in NVM CiMs. Conventional approaches include software runtime encryption [13], [14] and hardware data encryption such as AES [15], [16]. Despite the wide usage of these methods, software encryption still holds an attack window during algorithm execution, while AES-based encryption is incompatible with in-memory computing. Besides, both approaches incur significant performance and energy overheads, making them less feasible for edge devices.

Beyond the conventional approaches, recent works have explored the use of *in-situ* XOR cipher for lightweight encryption [17]–[24]. Through the subtle combination of inputs and keys, the weight decryption can be performed simultaneously with computing, achieving ultra-low cost and secure storage. However, the security of this method is premised on the secure generation and delivery of the keys, which is underexplored in prior designs. While physical unclonable functions (PUFs), which generate unique and unpredictable responses to given challenges, can be employed for key generation [20]–[24], the attacks against PUFs [25]–[27] and key delivery via register chains [28] still pose threats to the system, as shown in Fig. 1(b). Furthermore, the commonly used complementary ciphertext storage degrades both density and efficiency.

To cope with this challenge, we highlight the necessity of viewing PUF and CiM as inseparable parts of the system and propose **PUFiM**, a robust and efficient ferroelectric FET (FeFET)-based security solution merging PUF with CiM. As shown in Fig. 1(c), the core insight is to synergize PUF with CiM in a unified array to enable local key generation, decryption and computation with comprehensive security,

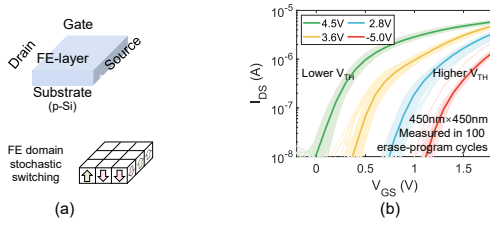


Fig. 2. FeFET: (a) Structure. (b) I_{DS} - V_{GS} curves with C2C variations [29].

where multi-level cell (MLC) FeFETs are exploited to support efficient PUF and weight hybrid storage and computation.

The highlighted contributions are summarized as follows:

- To the best of our knowledge, PUFiM is the **first** edge security solution that fuses PUF and CiM into one array for comprehensive and area-efficient protection.
- The startup value-based obfuscation and nonlinearity-enhancing post-processing are proposed to secure PUF.
- A significance-aware weight mapping scheme is proposed to improve CiM energy efficiency and security further.
- Extensive experiments are performed, and optimal configurations are explored. Results show that PUFiM reduces the inference accuracy by $>60\%$ under 95% key leakage, and achieves $>9.7\times$ compute density and $>1.2\times$ energy efficiency improvement compared with the state-of-the-art SRAM/NVM secure CiMs [18]–[21].

II. BACKGROUND AND RELATED WORKS

This section introduces the FeFET basics, illustrates the attack model and analyzes the bottlenecks of related works.

A. FeFET Basics

FeFET is a MOSFET with the insertion of a ferroelectric (FE) layer in the gate stack, as shown in Fig. 2(a) [29]–[31]. Reports have shown the fabrication of FeFET arrays at 28nm bulk [29] and 22nm FDSOI [31] technology node. Recent research has also reported the large-scale 3D integration of FeFETs [32] and the FeFETs with unlimited endurance [33].

The FE layer consists of multiple independent FE domains whose polarization states collectively affect the FeFET threshold voltage (V_{TH}). The multi-domain structure of the FE layer leads to two intriguing features [30] shown in Fig. 2(b): (i) By applying different write voltages (V_W) to the FeFET gate, the polarization state of the FE layer can be partially switched to achieve MLC storage; (ii) The polarization switching of a single FE domain is an abrupt and stochastic process due to the random polarization nucleation event, enabling FeFETs with significant cycle-to-cycle (C2C) variations.

Inspired by the two features, we leverage the FeFET MLC capability to achieve high-density hybrid storage and computation of the PUF and weight bits. Meanwhile, the FeFET C2C variation is exploited as the PUF entropy source by comparing the V_{TH} obtained from two write cycles of the same device.

B. Attack Model

The attacker is assumed to have access to an edge AI chip with an integrated secure CiM, which can be obtained from the market and used for inference as an authorized user.

The communication channel between the manufacturer or the model vendor and the chip can be monitored. Additionally, the attacker can carry out physical attacks to probe the weights of the DNN model stored in the NVM cells and the keys stored in the register chain, as demonstrated in [28], [34]. The attacker also has complete knowledge of the hardware implementation. However, due to the advancements in technology nodes, directly probing the signals within the secure CiM array is practically infeasible [35]. Therefore, the internal signals, including those related to the computation and key operations, are assumed to be secure from such attacks.

C. Related Works

To thwart model leakage with tolerable overheads, several lightweight secure CiM designs have been explored. A common approach involves pre-storing the encrypted weights in complementary cells with dual word lines (WLs), which allows the correct WL to be selected based on the key for *in-situ* decryption. Representative works include XOR-CiM [20], Secure-RRAM [17], SRA [18] and NAND FeFET CiM [19]. However, these designs double the memory cells and incur substantial area overheads. [36] eliminates the additional cells using a novel device, though its reliability and scalability remain challenges. Moreover, the above designs consider CiM a separate module instead of a system component, neglecting key generation and delivery as the basis for CiM security.

For secure key generation, PUF is employed as security primitives, exploiting entropy sources such as manufacturing variations to produce unique challenge-response pairs (CRPs). However, the weak PUFs utilized in [20]–[22] provide insufficient CRP space for authentication, thus vulnerable to attacks on communication channels [26]. In contrast, [23], [24] implement strong PUFs based on in-memory computing but face challenges from modeling attacks [25] and NVM data theft [34]. Furthermore, the separate implementations of PUF and CiM remain prone to key delivery attacks [28] and limit the opportunity for system performance co-optimization. Therefore, PUFiM is motivated to synergize PUF with CiM in one array for comprehensive security and improved efficiency.

III. PUFiM OVERVIEW

Fig. 3(a) illustrates the complete workflow of PUFiM from PUF enrollment to edge inference. The novel merging of PUF and CiM, combined with other highlighted techniques, makes PUFiM behave as a black box for the attacker and ensures robust and efficient protection throughout the workflow:

(i) **PUF enrollment with obfuscation.** After the PUFiM chip is manufactured in a trusted environment, the PUF bits are generated by exploiting the FeFET C2C variations and written back into the array. Based on the random startup values from the peripherals, a subset of the PUF bits is selected and activated for CRP generation, providing obfuscation for the nonvolatile PUF bits. These startup values and CRPs are subsequently stored in the cloud as the device fingerprint.

(ii) **Secure PUF authentication.** Before communication, PUFiM sends challenges to the cloud and verifies between the

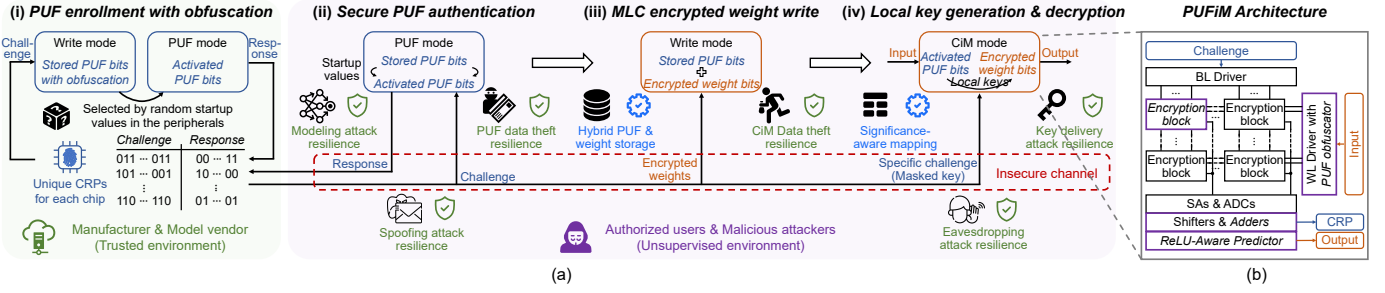


Fig. 3. Overview of PUFiM: (a) Complete workflow with security and efficiency enhancement approaches. (b) Overall architecture with blockwise structure.

received and self-generated responses for authentication. This prevents the *spoofing attack* against device identity. Besides, the proposed obfuscation and post-processing techniques provide resilience to the *PUF data theft* and *modeling attack*.

(iii) **MLC encrypted weight write.** Once authenticated, the model vendor encrypts the weights using the stored CRPs and offloads the encrypted weights to the edge to thwart *CiM data theft*. These weights are written to the MLC FeFETs based on the significance-aware mapping scheme and co-exist with the PUF bits, enabling high-density and secure hybrid storage.

(iv) **Local key generation, decryption and computation.** For edge inference, PUFiM acquires a specific challenge from the cloud and computes it with the activated PUF bits to generate local keys for each encryption block. These internal keys, distinct from the publicly accessible CRPs, are secure and can be retained until power-down. The local keys decrypt the weights in the block for computation. By merging PUF with CiM, the challenge (as a masked key) and the local keys protect PUFiM from *eavesdropping* and *key delivery attacks*.

Fig. 3(b) shows the PUFiM architecture. Challenges and DNN inputs are applied vertically and horizontally, respectively. During inference, multi-bit inputs are sequentially processed cycle by cycle, with the outputs accumulated using shifters & adders. Further summation of the outputs produces the final feature map. The encryption block with hybrid storage and computation is the core of PUFiM to enable the synergization of PUF and CiM. The startup value-based obfuscation and nonlinearity-enhancing post-processing are facilitated by the WL driver and the adders, while the significance-aware weight mapping improves efficiency via the ReLU-aware predictor.

IV. DESIGN DETAILS

This section details the four highlighted techniques of PUFiM, which synergizes PUF with CiM and significantly improves security, area efficiency and energy efficiency.

A. MLC Hybrid Storage and Computation

A key challenge in merging PUF and CiM lies in hybridly storing PUF and weight bits without affecting their respective computations. This becomes even more challenging with MLC, where the two bits stored in one device can cause interference, and the FeFET variations degrade the computing reliability. Therefore, PUFiM introduces a novel scheme to achieve both **high-density storage** and **reliable computation**.

Fig. 4(a) shows the proposed encoding and storage scheme. The stored bits are encoded into FeFET states with Gray code

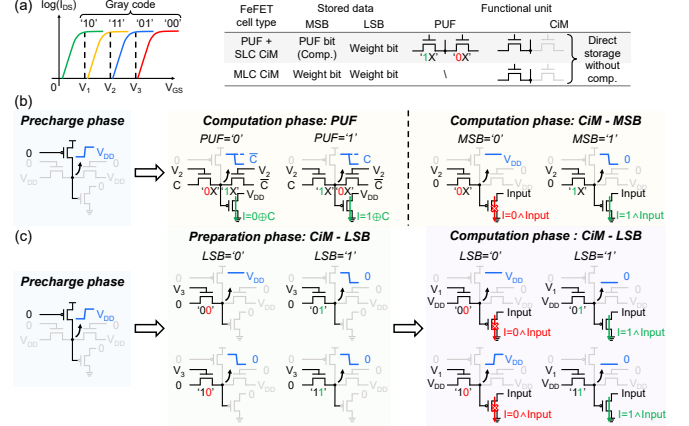


Fig. 4. MLC hybrid storage and computation: (a) Encoding and storage scheme. (b) MSB computation (PUF and CiM). (c) LSB computation (CiM).

and read out with $V_{1/2/3}$. For hybrid storage, FeFET cells are categorized into two types: (i) Some cells store weight bits in the LSB and complementary PUF bits in the MSB, with two such cells forming a PUF functional unit. (ii) Other cells store weight bits in both the MSB and the LSB. Note that all weight bits are stored directly and can be operated in one cell.

Fig. 4(b) shows the computation scheme for the MSB, which begins with a precharge phase. For PUF computation, V_2 is applied to the gates of two FeFETs with complementary PUF bits, and the challenge is input complementarily from the drains. This setup makes the PMOS on/off state consistent with the XOR result of the PUF and challenge bits. For CiM computation, the FeFETs storing '1' in the MSB discharge the internal node at V_2 , activating the PMOS if the input is VDD.

Fig. 4(c) shows the computation scheme for the LSB. The preparation phase sets the FeFET gate voltage to V_3 , discharging the internal nodes of all cells to GND except those storing '00'. Then V_1 is applied, and the FeFET drain is raised to VDD, which charges the '10' cells to switch off the PMOS. Consequently, only cells storing '1' in the LSB output currents.

The above method enables the hybrid storage of PUF and CiM while supporting their respective computations (XOR and AND). The FeFET variations are mitigated by converting the readout results into PMOS currents. The related evaluation of the PUFiM reliability is presented in Section V-B.

B. Local Key Generation, Decryption and Computation

Based on the proposed MLC hybrid storage and computation method, the encryption block circuit and operation mechanism are illustrated in Fig. 5 and detailed below:

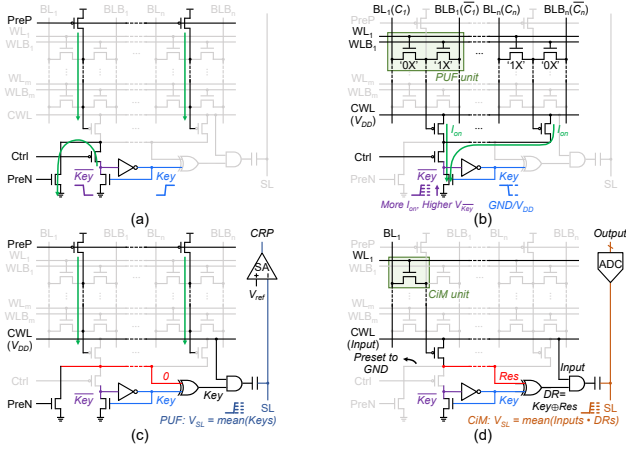


Fig. 5. Encryption block circuit and operation mechanism: (a) Precharge. (b) Local key generation. (c) PUF mode. (d) CiM mode.

(i) Precharge. The internal nodes in the FeFET array and the inverter output are precharged to VDD.

(ii) Local key generation. The challenge vector is input via BLs and BLBs to perform XOR computation with one row of PUF units. The resulting output currents are accumulated, affecting the voltage division at the inverter input. Once the voltage exceeds the inversion threshold, the inverter output (i.e., the key) switches to GND and shuts down the NMOS loop to lock the result. Otherwise, the output remains at VDD.

(iii) PUF-mode CRP generation. In PUF mode, the generated key is delivered to the capacitor connected to the SL. The capacitive coupling across encryption blocks averages the keys, and the obtained SL voltage is compared with a reference voltage by the sense amplifier (SA) to obtain the CRP. Note that this CRP requires post-processing for the final output.

(iv) CiM-mode decryption and computation. In CiM mode, one cell per block is activated to convert the weight bit into the voltage at Res , which is XORed with the generated local key for decryption. The decrypted result is subsequently ANDed with the input from the CWL for charge-domain computing.

The proposed encryption block generates and utilizes keys locally, effectively **preventing key delivery attacks**. The key generation requires only **one row of activated PUF bits** in each block without introducing heavy costs. The charge-domain computing offers energy-efficient and reliable computation, while the CMOS circuits can be shared by the FeFET array to reduce the area overheads. The block size configuration is analyzed and optimized in Section V-B.

C. PUF Obfuscation and Post-Processing

Despite the robust security of keys and weights in the PUFiM encryption block, the FeFET-based PUF remains vulnerable to probing and modeling attacks. **To thwart these threats**, obfuscation and post-processing techniques are proposed.

Fig. 6(a) illustrates the startup value-based obfuscation. Upon power-on, the bistable circuits output random startup values due to process variations, which are used to activate one of the two rows of PUF bits in a block. Therefore, the PUF bits stored in the nonvolatile FeFETs are obfuscated. However, some bistable circuits may be less robust and output unstable

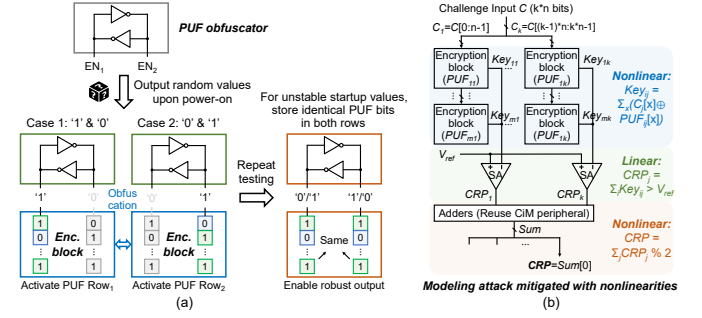


Fig. 6. PUF enhancement techniques. (a) Obfuscation. (b) Post-processing.

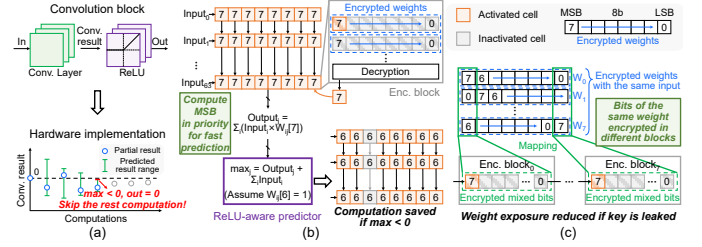


Fig. 7. Significance-aware weight mapping. (a) Insight of ReLU prediction. (b) Efficiency enhancement mapping. (c) Security enhancement mapping.

values. These circuits can be identified in advance, and the two PUF rows can be stored identically to enhance reliability at the cost of potential key leakage, as evaluated in Section V-D.

Fig. 6(b) shows the nonlinearity-enhancing post-processing with the complete CRP generation workflow. The XOR operations involved in key generation form the first nonlinear layer. Then m keys of the same column are aggregated in a linear approach. During post-processing, the adders designed for accumulating CiM outputs are **reused** to sum the results from k columns, with the LSB of the sum serving as the final CRP output. This additional nonlinearity increases the complexity of modeling the PUF without introducing additional area, and the improved resilience is shown in Section V-C.

D. Significance-Aware Weight Mapping

To further explore the potential of the PUFiM architecture based on encryption blocks, a significance-aware weight mapping scheme consisting of two sub-schemes is proposed.

Fig. 7(a) and (b) show the insight of the efficiency enhancement mapping. In convolution computation, the ReLU output would be 0 once the maximum possible value of the partial result is less than 0. As a result, the remaining computations can be skipped. Based on this observation, PUFiM maps all bits of one weight into the same encryption block and prioritizes the MSB computations in all blocks to accelerate prediction. If the predicted maximum output of any column is below 0, the corresponding blocks can be idled to save energy.

Fig. 7(c) shows the security enhancement mapping. The core idea is to shuffle the bits of the weights with the same input and map them into different encryption blocks without affecting the computations. This ensures that bits of the same weight are encrypted by different keys, significantly reducing the weight exposure caused by key leakage from any single block.

Both sub-schemes can be applied simultaneously to improve **energy efficiency and security** as shown in Section V-D.

TABLE I
PUFiM PARAMETERS

Unit	Spec	
FeFET array	Array size ¹	1024×1024 FeFETs
		0.22fF/ μm parasitic capacitance
	FeFET dimension ²	Exp: 450nm×450nm Sim: 200nm×100nm
	Write voltage	-5.0V/2.2V/3.6V/4.5V, 500ns
	Read voltage	$V_1/V_2/V_3 = 0.3\text{V}/0.95\text{V}/1.2\text{V}$
	V_{TH} variation	130mV/85mV/176mV/110mV
ADCs & SAs	Capacitor	1fF MOM, $\sigma = 0.8\%$, $2.74\mu\text{m}^2$
	5-bit SAR ADC, VSA, 1 per column of blocks	

¹ The encryption block size is 8×10 based on the optimizations in Section V-B, and two rows per block store PUF bits in the MSBs.

² The FeFET layout in Fig. 9 is close to the fabricated ones.

V. EVALUATION AND DISCUSSION

A. Experimental Setup

Hardware-level evaluation: To comprehensively capture FeFET characteristics, we utilize the experimental measurement results of 9×7 FeFET arrays fabricated at 28nm node to extract the C2C variation, MLC storage, and device non-idealities for accurate evaluation. The area, latency and energy consumption of FeFETs are simulated based on a widely used FeFET model [37] using the parameters from the fabricated FeFET arrays. 5-bit SAR ADCs and voltage SAs (VSAs) are designed, and digital peripherals are synthesized at 28nm node. The array size is set to 1024×1024 FeFETs with the parasitic capacitance considered. Table I lists the detailed information.

Software-level evaluation: Five powerful modeling attacks are performed, i.e., LR, MLP, SVM, XGBoost and CMA-ES, which have shown success against various PUF designs [25], [26], [38]. The inference accuracy is evaluated on VGG8 for CIFAR10 and ResNet18 for CIFAR100, covering different dataset scales and network complexity. We use PyTorch for quantization-aware training to obtain the model with 8-bit input and 8-bit weight. The inference process is modified to include the crypto operations and non-idealities from hardware.

Baselines: The modeling attack resilience is benchmarked with the arbiter PUF with and without 6-XOR structure [39] and the state-of-the-art FeFET PUF [23]. For secure CiM evaluation, four state-of-the-art designs based on SRAM [20], RRAM [18], STT-MRAM [21] and FeFET [19] are compared. The technology node of all baselines is scaled to 28nm using DeepScale [40] for accurate and apple-to-apple comparisons.

B. Circuit-level Evaluation and Optimization

Fig. 8 evaluates the reliability of the encryption block. As shown in Fig. 8(a), considering FeFET variations, the outputs of PUF units still exhibit tight distributions primarily induced by the CMOS variation, while the outputs of CiM units show an error rate of $<2\%$ due to the FeFET state overlap. This is attributed to the proposed MLC hybrid computation method, which converts FeFET readout results to CMOS currents. For key generation, the accumulation of output currents is sensed by the inverter-NMOS loop with $<5\%$ error rate, facilitating

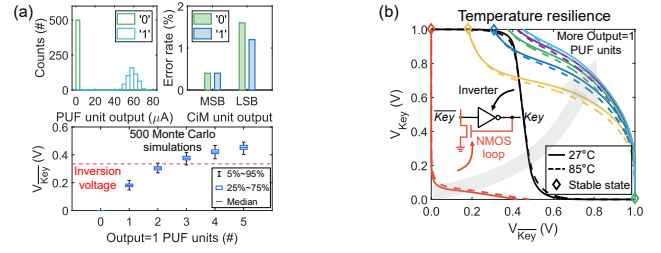


Fig. 8. Encryption block reliability: (a) Distributions and error rates of the PUF and CiM unit outputs. (b) Temperature resilience of key generation.

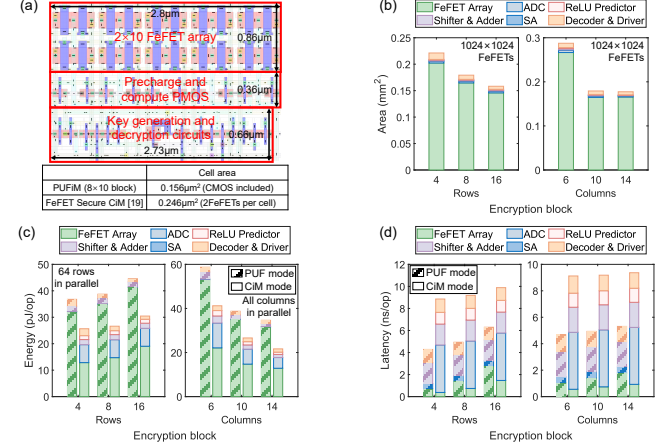


Fig. 9. Encryption block performance: (a) Layout of 2×10 block and cell area comparison with prior works [19]. (b) Area. (c) Energy. (d) Latency.

high randomness and uniqueness of the keys. Fig. 8(b) further shows the temperature resilience of the sensing circuits. Thanks to the voltage division between transistors in series, the intersection of the inverter-NMOS loop in-out curves, representing stable outputs, shifts negligibly as temperature varies from 27°C to 85°C, which indicates robust resilience.

Fig. 9(a) shows the layout of a 2×10 encryption block based on 28nm design rules. The width of the CMOS circuits perfectly matches the FeFETs, and the capacitor ($2.74\mu\text{m}^2$) can be placed on top. Consequently, PUFiM achieves $0.156\mu\text{m}^2$ per cell with CMOS included, which is $1.6 \times$ lower than the complementary storage in [19] with the same device size.

Fig. 9(b-d) evaluate the PUFiM area, energy and latency in both PUF and CiM modes with different encryption block sizes. The FeFET array occupies most of the area as the blocks share the peripherals, and the resulting parasitic capacitance further contributes to a high energy proportion. Compared with the CiM mode, the PUF mode consumes more energy due to the power-hungry current summation required for key generation but shows less latency with the ADCs not involved.

For the impact of the encryption block size, increasing rows and columns enables the CMOS circuits to be shared by more FeFETs and reduces the area. Besides, increasing the rows within a block, while maintaining the same row parallelism, heavies the precharging burden and results in more energy and latency. While more columns have a similar effect on precharging, their impact on energy is covered by the reduction in column parallelism. With the above observations, we select 8×10 as the optimized encryption block size for PUFiM.

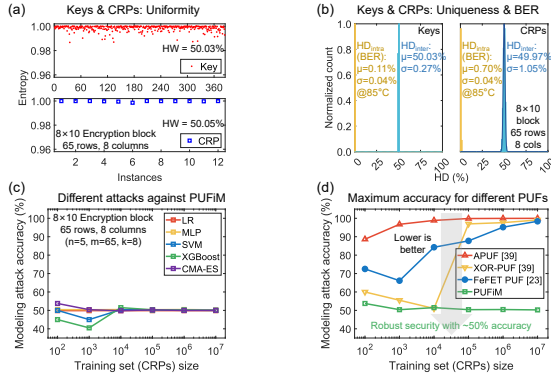


Fig. 10. PUF mode evaluation: (a) Uniformity. (b) Uniqueness and BER. (c) Modeling attack resilience of PUFiM. (d) Comparison with existing PUFs.

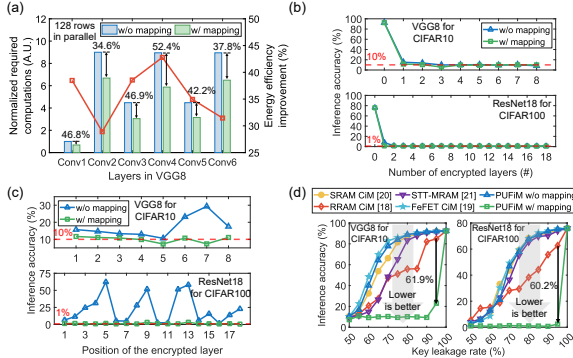


Fig. 11. CiM mode evaluation: (a) Energy efficiency improvement of the significance-aware weight mapping. (b) Encryption security performance. (c) Impact of encrypted layer position on security. (d) Key leakage resilience.

C. PUF Performance and Security Evaluation

Fig. 10(a) and (b) show the basic PUF metrics of PUFiM. Both the keys (within encryption blocks) and CRPs (after post-processing) show excellent $\sim 50\%$ uniformity and uniqueness, and achieve an ultra-low $<0.70\%$ bit error rate (BER) at 85°C . This is attributed to the proposed reliable computation method and the use of FeFET C2C variations as the entropy source.

Fig. 10(c) shows the PUFiM modeling attack resilience across multiple attack models. Thanks to the nonlinearity provided by the proposed post-processing technique, PUFiM achieves a maximum accuracy of 50.26% (close to random guessing) even with a training set size of 10M, which is almost infeasible for the attacker to obtain. When applying these attacks to various PUFs, PUFiM shows significantly enhanced security compared with the vulnerable baselines.

D. CiM Performance and Security Evaluation

Fig. 11(a) shows the benefits of the proposed significance-aware mapping in VGG8. By accelerating the ReLU prediction convergence, the required computations are reduced with the related blocks idled, improving the PUFiM energy efficiency.

Fig. 11(b-d) evaluate the security performance of PUFiM inference. Encrypting a single layer drastically reduces accuracy to unusable levels, with the position of the layer affecting the accuracy drop. The proposed weight mapping scheme ensures consistently low accuracy levels close to random guessing. As shown in Fig. 11(d), thanks to the finer-grained encryption and the proposed mapping scheme, PUFiM achieves much higher

TABLE II
COMPARISON BETWEEN PUFiM AND SOTA SECURE CiMs

	[20]	[18]	[21]	[19]	PUFiM
Cell type	SRAM	RRAM	STT-MRAM	FeFET	FeFET
Technology	28nm	32nm	22nm	28nm	28nm
Bit/cell	1	1	1	2	2
PUF support	Weak	No	Weak	No	Strong ¹
Key delivery vulnerability	Yes	Yes	Yes	Yes	No ²
Key leakage vulnerability ³	74.88%	62.65%	73.88%	75.59%	2.46%
Storage density (Mb/mm ²)	0.05	6.85	0.73	0.61	10.52 ($>1.5\times$)
Compute density ⁴ (TOPS/mm ²)	0.004	0.046	0.044	0.011	0.444 ($>9.7\times$)
Energy efficiency ⁴ (TOPS/W)	2.86	0.33	12.24	8.08	14.75 ($>1.2\times$)

¹ PUFiM also achieves resilience against modeling attack and data theft on NVM PUFs.

² The keys are generated and used internally without the delivery via register chains.

³ Inference accuracy of ResNet18 for CIFAR100 with 95% key leakage.

⁴ Normalized to 8-bit input, 8-bit weight precision with 50% sparsity.

security compared with the prior secure CiMs, lowering the accuracy by $>60\%$ under 95% key leakage on both models.

E. Macro-Level Comparison with SOTA Designs

Table II summarizes the macro-level comparison between PUFiM and the state-of-the-art secure CiMs [18]–[21]. **From a security perspective**, the merging of PUF and CiM eliminates the key delivery vulnerabilities, while the proposed PUF enhancement techniques and the significance-aware mapping further improve the resilience against PUF attacks and key leakage. This comprehensive security solution surpasses all the baselines in robustness for edge AI systems. **For area and energy efficiency**, thanks to the proposed MLC hybrid storage and computation, PUFiM achieves $>1.5\times$ storage density and $>9.7\times$ compute density compared with the baselines. Besides, the compact merging saves the peripherals, and the mapping scheme reduces the required computation, collectively achieving $>1.2\times$ energy efficiency improvement.

F. More Discussions

PUFiM is designed for edge inference, where infrequent weight updates mitigate FeFET endurance concerns. The reliability of MLC FeFETs can be further improved by write-and-verify [41]. Besides, resilience against chosen input attacks (i.e., collecting input-output pairs for retraining) and side-channel attacks can be improved through enhanced DNN model robustness [42] and masking techniques [43].

VI. CONCLUSION

This paper proposes PUFiM, a FeFET-based edge security solution with comprehensively robust security and significant efficiency improvement. The key highlight is the merging of PUF and CiM through MLC hybrid storage and computation, which enables highly secure local key generation and decryption. The obfuscation and post-processing techniques are proposed to enhance PUF security, and a significance-aware mapping scheme tailored for the PUFiM architecture is developed to improve CiM energy efficiency and security. Results show that PUFiM outperforms the state-of-the-art secure CiMs in security, area efficiency and energy efficiency.

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